

Metadata Agent

Every asset describes itself — from evidence, with its reasoning attached.

THE PROBLEM

Why this matters

Enterprise DAMs are full of assets nobody can find. Metadata is entered by hand at the end of a deadline, so it is skipped, inconsistent, or wrong — and every AI-generated adaptation multiplies the backlog. Worse, an adaptation silently inherits the rights of every source it came from, so a cropped-out licensor keeps claiming rights it no longer has. Manual cataloguing never catches up with the volume creative teams now produce.

An asset nobody can find is an asset nobody paid for.

What the agent does

1

Read the folder's contract

The agent inspects the metadata schema AEM applies to the folder: which fields exist, which are mandatory, and the only values a closed list will accept.

2

Learn how the team actually works

It samples comparable assets in the same folder and discovers the convention they already follow, rather than inventing its own vocabulary.

3

Look at the asset itself

A vision model describes what is genuinely in the pixels — prompted with the schema and the folder's own words, so it stays in the customer's taxonomy.

4

Inherit from the real lineage

For an adaptation it resolves every source through sourceUUID and inherits what those sources agree on, flagging where they conflict.

5

Delegate the rights question

Digital rights are never inferred. The agent asks the Rights Reconciliation Agent over A2A and takes only the rights that survived into the frame.

6

Decide, write, and show its working

Each field is populated, sent for human review, or left alone — and every value written carries the lanes, confidence and evidence behind it.

HOW IT DECIDES

Five lanes, weighed in one place

Each lane investigates independently. None of them can act alone — a resolver weighs them, and a lane is never trusted beyond its own remit.

Lane	Trust	What it contributes
Rights	1.00	Delegated to the Rights Reconciliation Agent over A2A. The only source of rights values.
Lineage	0.95	What an adaptation's source assets agree on. Lists union; disagreement is flagged, not averaged.
Folder schema	0.90	The contract AEM applies: which fields exist, which are required, which values are legal.
Siblings	0.80	The convention comparable assets already follow — discovered from evidence, not assumed.
Visual	0.75	A vision model, prompted with the schema and the folder's own vocabulary.

Above 0.75 the agent writes. Below it, or whenever two lanes genuinely disagree, a human decides — and sees why.

What ships first

- Live agent on Adobe App Builder, reachable as both an MCP server and an A2A endpoint, with a published Agent Card.
- Five independent evidence lanes — folder schema, sibling consensus, visual inference, lineage inheritance, and delegated rights.
- Confidence-weighted resolver: auto-populate above 0.75, human review below it, and always on genuine disagreement.
- Write-back to AEM Assets under a schema-derived allow-list, with full mda* provenance on every asset.
- Rights delegated to the Rights Reconciliation Agent over A2A — loopback today, a URL change to go remote.
- Running prototype against AEM as a Cloud Service, 155 automated tests green.

Where this goes

- Event-driven: the agent catalogues every asset the moment AEM finishes processing it — no human trigger at all.
- The DAM's taxonomy is learned continuously, so the agent's vocabulary tracks how the customer's language actually changes.
- A curator's correction is fed back as training signal, so the agent measurably improves per customer rather than staying static.
- A marketplace of evidence lanes — brand compliance, accessibility, product recognition — added without touching the resolver.
- The same reasoning enforced at download and publish, so no asset leaves the DAM under-described or over-claiming rights.

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THE VALUE

What changes

5

independent evidence lanes
weighed per field

0

rights fields ever inferred — always
delegated

100%

written values carry traceable
provenance

~70%

manual tagging effort removed
(target)

The first three are properties of the design, not projections. The fourth is the outcome we intend to validate with a customer folder.

WHY IT WINS

Our case

- It is agentic, not generative: the agent decides what it knows, what it must ask a human about, and what it must refuse to touch — then acts on that decision autonomously.
- It shows its working. Every value written records which lanes produced it and how confident they were, so a curator can audit any field months later.
- It knows the limits of its own authority — rights are delegated to a specialist agent, technical facts are left to AEM, and a curator's existing value is never overwritten.
- It is real today: a working agent on Adobe App Builder against AEM as a Cloud Service, with 155 tests, not a slideware concept.
- It is reusable by construction — it reads whatever schema the customer applied, so it fits the next DAM without a rewrite.
- Agent-to-agent delegation over open standards (A2A, MCP) makes it a citizen of the Adobe agentic ecosystem, not an island.

BUILT ON

The stack

Layer	What we use	Licence	Where it runs
Runtime	Adobe App Builder · Adobe I/O Runtime (Node.js)	Adobe	Adobe-hosted
Agent protocols	Agent Card · A2A · MCP	Open standards	Adobe-hosted
Platform	AEM Assets as a Cloud Service	Adobe	Adobe-hosted
AI vision	Claude / OpenAI-compatible, with fallback	Hosted, optional	Configurable
Agent interop	Rights Reconciliation Agent over A2A	Internal agent	Adobe-hosted
Testing	Jest — 155 automated tests, all passing	OSS	Local
Data	Synthetic assets — no customer data	N/A	Local

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Metadata Agent · Adobe · ACS / GDC

The Guardians